



| Revision Dates |  |          |
|----------------|--|----------|
| 1              |  | 08.15.24 |
| 2              |  | 01.01.25 |
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FRAMING NOTES: (SEC. 2308.9.10 & 2308.9.11)

1. NOTCHING OF STUDS IN BEARING PARTITIONS SHALL NOT EXCEED 25% OF THE WIDTH.

2. NOTCHING OF STUDS IN NONBEARING PARTITIONS SHALL NOT EXCEED 40% OF THE WIDTH.

3. BORED HOLES IN STUDS IN BEARING PARTITIONS SHALL NOT EXCEED 40% OF THE WIDTH.

4. BORED HOLES IN STUDS IN NONBEARING PARTITIONS SHALL NOT EXCEED 60% OF THE WIDTH, PROVIDED THE BORED STUDS ARE DOUBLED AND NO MORE THAN TWO SUCH SUCCESSIVE DOUBLED STUDS ARE SO BORED.

5. BORED HOLES MAY NOT BE LOCATED NEARER THAN 5/8" TO THE EDGE OF THE STUD AND MAY NOT BE LOCATED IN THE SAME SECTION OF THE STUD AS A CUT OR NOTCH.

6. RAFTERS SHALL BE NAILED TO ADJOINING CEILING JOISTS TO FORM A CONTINUOUS TIE ACROSS THE BUILDING. IF RAFTERS ARE NOT PARALLEL TO THE CEILING JOISTS, CROSS TIES SHALL BE PROVIDED DIRECTLY ABOVE THE CEILING JOISTS. TIES SHALL BE SPACED AT NOT MORE THAN 4'-0" O/C. PROVIDE A RAFTER TIE CONNECTION DETAIL IN ACCORDANCE WITH TABLE 2308.10.4.1. (SEC.2308.10.4.1)

7. NOTCHING AT THE ENDS OF RAFTERS OR CEILING JOISTS SHALL NOT EXCEED ONE-FOURTH THE DEPTH. NOTCHES IN THE TOP OR BOTTOM OF THE RAFTER OR CEILING JOIST SHALL NOT EXCEED ONE-SIXTH THE DEPTH AND SHALL NOT BE LOCATED IN THE MIDDLE ONE-THIRD OF THE SPAN, EXCEPT THAT A NOTCH NOT EXCEEDING ONE-THIRD OF THE DEPTH IS PERMITTED IN THE TOP OF THE RAFTER OR CEILING JOIST NOT FURTHER FROM THE FACE OF THE SUPPORT THAN THE DEPTH OF THE MEMBER.

SMOKE ALARM NOTES:

- ALL SMOKE ALARMS SHALL BE LISTED IN ACCORDANCE WITH UL 217 AND INSTALLED IN ACCORDANCE WITH THE PROVISIONS OF THE GOVERNING CRC AND THE HOUSEHOLD FIRE WARNING EQUIPMENT PROVISIONS OF NFPA 72. SYSTEMS AND COMPONENTS SHALL BE CALIFORNIA STATE FIRE MARSHAL LISTED AND APPROVED IN ACCORDANCE WITH CCR, TITLE 19, DIVISION 1 FOR THE PURPOSE FOR WHICH THEY ARE INSTALLED. [SEC. R314.1]."
- SMOKE ALARMS SHALL RECEIVE THEIR PRIMARY POWER FROM THE BUILDING WIRING PROVIDED THAT SUCH WIRING IS SERVED FROM A COMMERCIAL SOURCE AND SHALL BE EQUIPPED WITH A BATTERY BACK-UP. SMOKE ALARMS WITH INTEGRAL STROBES THAT ARE NOT EQUIPPED WITH BATTERY BACKUP SHALL BE CONNECTED TO AN EMERGENCY ELECTRICAL SYSTEM. SMOKE ALARMS SHALL EMIT A SIGNAL WHEN THE BATTERIES ARE LOW. WIRING SHALL BE PERMANENT AND WITHOUT A DISCONNECTING SWITCH OTHER THAN AS REQUIRED FOR OVERCURRENT PROTECTION. [SEC. R314.4]
- WHERE MORE THAN ONE SMOKE ALARM IS REQUIRED TO BE INSTALLED WITHIN AN INDIVIDUAL DWELLING UNIT THE ALARM DEVICES SHALL BE INTERCONNECTED IN SUCH A MANNER THAT THE ACTIVATION OF ONE ALARM WILL ACTIVATE ALL OF THE ALARMS IN THE INDIVIDUAL UNIT. [SEC. R314.3].

BATHTUB NOTES:

BATHTUB AND SHOWER FLOORS AND WALLS ABOVE BATHTUBS WITH INSTALLED SHOWER HEADS AND IN SHOWER COMPARTMENTS SHALL BE FINISHED WITH A NONABSORBENT SURFACE. SUCH WALL SURFACES SHALL EXTEND TO A HEIGHT OF NOT LESS THAN 6 FEET ABOVE THE FLOOR. [SEC. R307.2].

NAILING SCHEDULE: (From Table 2304.9.1)

1. JOIST TO SILL OR GIRDER, TOENAIL 3- 8D

2. BRIDGING TO JOIST, TOENAIL EACH END 2- 8D

3. SOLE PLATE TO JOIST OR BLOCKING, FACE-NAIL 16D @ 16" O.C.

4. SOLE PLATE TO JOIST OR BLOCKING, AT BRACED WALL PANELS 3-16D PER 16"

5. TOP PLATE TO STUD, END-NAIL 2-16D

6. STUD TO SOLE PLATE 4-8D TOE NAILED, OR 2-16D END-NAILED

7. DOUBLE STUDS, FACE-NAILED 16D @ 24" O.C.

8. DOUBLE TOP PLATES, FACE-NAIL 16D @ 16" O.C.

9. DOUBLE TOP PLATES, LAP SPLICED 8-16D

10. BLOCKING BETWEEN JOISTS OR RAFTERS TO TOP PLATE, TOENAIL 3-8D

11. RIM JOIST TO TOP PLATE, TOENAIL 8D @ 6" O.C.

12. CEILING JOIST TO TOP PLATE, 3-8D TOENAILS

13. CONTINUOUS HEADER, TWO PIECES 16D @ 16" O.C. ALONG EACH EDGE

14. CONTINUOUS HEADER TO STUD, TOENAIL 4-8D

15. CEILING JOISTS TO PLATE, TOENAIL 3-8D

16. CEILING JOISTS, LAPS OVER WALL, FACE NAIL 3-16D (SEE SHT. A4 FOR RAFTER TIE NAILING)

17. RAFTER TO PLATE, TOENAIL 3-8D

18. BUILT-UP CORNER STUDS 16D @ 24" O.C.

19. BUILT-UP GIRDER AND BEAMS 20D @ 32" O.C. FACE NAIL AT TOP & BOTTOM AND STAGGERED ON OPPOSITE SIDES, PLUS 2-20D FACE NAIL AT EACH END

20. COLLAR TIE TO RAFTER, FACE NAIL, 3-10D (SEE SHT. A4 FOR RAFTER TIE NAILING)

21. JACK RAFTER TO HIP, TOENAIL 3-1 OD, FACE NAIL 2-16D

22. ROOF RAFTER TO 2X RIDGE BEAM, TOENAIL OR FACE NAIL 3-16D

23. JOIST TO BAND JOIST, FACE NAIL, 3-16D

24. LEDGER STRIP, FACE NAIL3-16D

NOTE: USE COMMON NAILS, NOT BOX, COOLER ETC.

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|-----|---------------------|-----|---------------------|
| 20d | 4" X 0.192" DIA     | 10d | 3" X 0.148" DIA     |
| 16d | 3 1/2" X 0.162" DIA | 8d  | 2 1/2" X 0.131" DIA |
| 12d | 3 1/4" X 0.148" DIA | 6d  | 2" X 0.113" DIA     |

MACHINE NAILING IS PROHIBITED AT DOUBLE SHEATHED PLYWOOD WALLS OR OTHER APPLICATIONS WHERE THE PRESENCE OF "SHINERS" CAN NOT BE DETECTED BY VISUAL OBSERVATION.

**8. BOLTS:**

- ASTM A307, U.N.O.
- RETIGHTEN ALL BOLTS BEFORE CLOSING IN.
- UPSET THREADS ARE PROHIBITED

9. ALL BEAMS SHALL BE SEAT CUT FOR FULL BEARING.

**SPECIAL HAZARDS**

- APPROVED SEISMIC GAS SHUTOFF VALVE WILL BE INSTALLED ON THE FUEL GAS LINE ON THE DOWN STREAM SIDE OF THE UTILITY METER AND BE RIGIDLY CONNECTED TO THE EXTERIOR OF THE BUILDING OR STRUCTURE CONTAINING THE FUEL GAS PIPING.
- WATER HEATER MUST BE STRAPED TO WALL.(SEC.507.3 UPC)

**MISCELLANEOUS**

- PROVIDE LOW CONSUMPTION WATER CLOSETS FOR ALL NEW CONSTRUCTION. SHOWER HEADS AND TOILETS MUST BE ADAPTED FOR LOW WATER COMSUMPTION.
- PROVIDE 70 INCHES HIGH NON-ABSORBENT WALL ADJACENT TO SHOWER AND APPROVED SHATTER-RESISTANT MATERIALS FOR SHOWER ENCLOSURE.
- PROVIDE DOUBLE JOISTS UNDER PARALLEL BEARING PARTITIONS.
- RAFTER TIES SPACED 4FT.(MAX.) O.C. ARE REQUIRED IMMEDIATELY ABOVE CEILING JOISTS WHICH ARE NOT PARALLEL TO THE RAFTERS.

**FOOTNOTES:**

- COMMON NAILS SHALL BE USED U.N.O.
- COMMON OR DEFORMED SHANK.
- COMMON.
- DEFORMED SHANK.
- NAILS SPACED AT 6 INCHES ON CENTER AT EDGES, 12 INCHES AT INTERMEDIATE SUPPORTS EXCEPT SHEAR WALLS, REFER TO SECTION 2314A3. NAILS FOR WALL SHEATHING SHALL BE COMMON.
- CORROSION-RESISTANT SIDING OR CASING NAILS CONFORMING TO THE REQUIREMENT OF SECTION 2325A.1.
- FASTENERS SPACED 3 INCHES ON CENTER AT EXTERIOR EDGES AND 6 INCHES ON CENTER AT INTERMEDIATE SUPPORTS.
- CORROSION-RESISTANT ROOFING NAILS WITH 7/16-INCH-DIAMETER HEAD AND 1 1/2-INCH LENGTH FOR 1/2-INCH SHEATHING AND 1 3/4 INCH LENGTH FOR 25/32-INCH SHEATHING CONFORMING TO THE REQUIREMENTS OF SECTION 2325A.1.
- CORROSION-RESISTANT STAPLES WITH NOMINAL 7/16 INCH CROWN AND 1 1/8-INCH LENGTH FOR 1/2-INCH SHEATHING AND 1 1/2-INCH LENGTH FOR 25/32-INCH SHEATHING CONFORMING TO THE REQUIREMENTS OF SECTION 2325A.1.
- PANEL SUPPORTS AT 16" IF STRENGTH AXIS IN THE LONG DIRECTION OF THE PANEL, UNLESS OTHERWISE MARKED. CASTING OR FINISH NAILS SPACED 6" ON PANEL EDGES, 12" AT INTERMEDIATE SUPPORTS. SECTION 2325A.1.
- PANEL SUPPORTS AT 24". CASING OR FINISH NAILS SPACED 6" ON PANEL EDGES, 12" AT INTERMEDIATE SUPPORTS.
- WHEN POSSIBLE, NAILS DRIVEN PERPENDICULAR TO THE GRAIN SHALL BE USED INSTEAD OF TOENAILS.

**CONNECTIONS**

- 5/8" Ø ANCHOR BOLT 4" MAX. O.C., 12" FROM CORNER. 7" MIN. IN CONC. FOUNDATION ALL BOLT HOLES SHALL BE DRILL 1/32" TO 1/16" OVERSIZED.
- ANCHOR BOLTS INCLUDING HOLD DOWN BOLTS ANCHORS SHALL INSTALLED AT 1 3/4" EDGE DISTANCE. CORNER DISTANCE FROM EDGE OF STRAP TO CORNER OF CONC.FOUNDATION IS 1 3/4"
- PLATE WASHERS ARE REQUIRED FOR ALL HOLD DOWNS OR ANCHOR BOLTS W./ 3"x3"x.229" METAL WASHER,
- PROVIDE 2 #4 REBARS AT TOP AND BOTTOM, WITH 3" CLEARANCE FROM EARTH. 6" MIN. EMBEDING INTO EXISTING FOOTING W/ AN APPROVED GLUE OR EQUIVALENT. 60° L, 24" INTO NEW CONCRETE FOOTING.
- PROVIDE #4 REBARS 24" O.C. 6" MIN. EMBEDED TO (E.)FOUNDATION W/ APPROVED GLUE OR EQUIVALENT, 18" L 60° INTO NEW FOUNDATION
- TIE ALL NEW WALL TO EXIST. WALL AT PLATE WITH MINIMUM MST-37 INSTALLED IN APPROVED MANNER OR EQUIVALENT

**GRADING**

- BUILDING SITTING: DRAINAGE 1/4":12" SLOPE, 3' AWAY FROM THE STRUCTURE, DRAINING FROM HIGH POINT, FROM THE BACK TO PUBLIC STREET. (SEE GRADING PLAN)

**FINISH**

- PROVIDE 6" WEEP SCREED FOR STUCCO AT THE FOUNDATION PLATE LINE A MINIMUM OF 4" ABOVE GRADE, 3 1/2" VERTICAL FLANGE WITH TOP VERTICAL FLANGE EDGE AT THE TOP.
- RECEPTACLES SHALL BE12" O.C. MAX. RULE FOR WALL. AND INSTALL 2' OR MORE WALL SPACE. ALL RECEPTACLES IN BATHROOM SHALL HAVE G.F.I.
- WALL COVERING SHALL BE CEMENT PLASTER, CERAMIC TILES, OR APPROVED MATERIALS TO 70 INCHES ABOVE DRAIN AT SHOWER OR TUBS WITH SHOWER. MATERIALS ORTHER THAN STRUCTURAL ELEMENTS TO BE MOITURE RESISTANT. WOOD FRAMING SHALL BE PROTECTD WITH APPROVED WATER PROOF PAPER. GLASS ENCLOSURE DOORS AND PANELS MUST LABELED CATEGORY II. SWING DOOR OUTWARD. NET AREA OF SHOWER RECEPTOR SHALL BE NOT LESS THAN 1024 SQ/IN OF FLOOR AREA AND ENCOMPASS 30" DIA. CIRCLE.

**GENERAL:**

- HOLES AND OPENINGS THROUGH WALLS AND FLOORS FOR DUCTS, PIPING AND VENTILATION SHALL BE CHECKED BY THE CONTRACTOR WHO SHALL VERIFY SIZES AND LOCATION OF SUCH HOLES OR OPENINGS WITH THE MECHANICAL, PLUMBING, AND ELECTRICAL DRAWINGS AND THESE SUB-CONTRACTORS. DETAILED OR APPROVED BY THE ENGINEER.
- NO PIPES OR DUCTS SHALL BE PLACED IN SLABS OR WALLS UNLESS SPECIFICALLY
- CONTRACTOR SHALL VERIFY ALL DIMENSIONS AT JOB SITE BEFORE COMMENCING WORK AND SHALL REPORT ANY DISCREPANCIES TO THE ENGINEER AND DESIGNER
- OMISSIONS OR CONFLICTS BETWEEN VARIOUS ELEMENTS OF THE DRAWINGS, NOTES, AND DETAILS SHALL BE BROUGHT TO THE ATTENTION OF THE DESIGNER AND RESOLVED BEFORE PROCEEDING WITH THE WORK.
- ATTIC ACCESS 22"x30" MIN. W./ OVER 30" HEADROOM.
- CRAWLSPACE VENTILATION SHALL BE NOT LESS THAN 1/150 OF UNDER FLOOR AREA
- UPER GROUND IS REQUIRED AT NEW ELECTRICAL SERVICE
- IAPMO APPROVAL REQUIRED FOR ONE PIECE LAVATORY

**DESIGN BASIS:**

- APPLICABLE CODE: CALIFORNIA CODE OF REGULATION, CBC 2010 EDITION, (CCR), TITLE 24, PART 2, ALSO REFERRED TO AS CALIFORNIA BUILDING CODE (CBC)

**MATERIALS:**

**A. CONCRETE**

- REINFORCING STEEL:
  - BARS: ASTM A615, GRADE 60.
  - ALL CONCRETE SHALL BE REINFORCED UNLESS SPECIFICALLY MARKED 'NOT REINFORCED'.
- CONCRETE CLASSES: SEE SPECIFICATIONS FOR REQUIREMENTS.

| CLASS | USE                          | WT     | STRENGTH          |
|-------|------------------------------|--------|-------------------|
| A.    | FOUNDATIONS & SLABS-ON-GRADE | 45 PCF | 2500 PSI          |
|       |                              |        | USE TYPE V CEMENT |

- MINIMUM CONCRETE COVER FOR REINFORCING STEEL:
  - SURFACES PLACED AGAINST EARTH 3"
  - FORMED SURFACES BELOW GRADE 2"
  - SURFACES EXPOSED TO WEATHER 2"
  - SLABS AND WALLS NOT 1" EXPOSED TO WEATHER

**B. WOOD**

- FRAMING LUMBER - DOUGLAS FIR; LARCH U.NO.:
  - 2x, AND/OR 4x JOISTS AND RAFTERS: NO. 1
  - 4x, 6x, POSTS, BEAMS, SILLS, PLATES AND HEADERS: NO. 1
  - 2x, STUDS, SILLS, BLOCKS, LIGHT FRAMING AND MISC: NO. 1
  - ALL LUMBER IN CONTACT WITH CONCRETE TO BE PRESSURE TREATED.
  - THE MOISTURE CONTENT OF ALL LUMBER 4x OR GREATER SHALL BE VERIFIED BY PROJECT INSPECTOR AT THE TIME OF FRAMING.
- MOISTURE CONTENT SHALL NOT EXCEED 19% (i.e. HEADERS, TOP PLATES, SILLS AND STUDS. 2x MEMBERS SHALL BE STAMPED "S-DRY". RMT PRE-MANUFACTURED MEMBERS PER DSA PRODUCT APPROVAL NO. PA-109 IS AN ACCEPTABLE ALTERNATE TO SOLID SAWN LUMBER.
- PLYWOOD SHEATHING:
  - ROOFING SHEATHING: 1/2 INCH. APA RATED STRUCT 1, EXPOSURE 1, 5 PLY.
  - WALL SHEATHING: 15/32 INCH APA RATED STRUCT. I, EXPOSURE 1, 5 PLY.
  - SHEAR WALL SEE DETAIL
  - PROVIDE DOUBLE JOISTS UNDER PARALLEL BEARING PARTITIONS.
  - RAFTER TIES SPACED 4FT.(MAX.) O.C. ARE REQUIRED IMMEDIATELY ABOVE CEILING JOISTS WHICH ARE NOT PARALLEL TO THE RAFTERS.
- FRAMING HARDWARE: AS MANUFACTURED BY SIMPSON.
- WOOD FRAMING MEMBERS: GRADE AND SPECIES OF ALL LUMBER "MUST BE GRADE MARKED".
- NAILS:
  - COMMON WIRE GAGE
  - NAILING TO CONFORM TO CBC TABLE 2304.9.1
  - PRE-DRILL FOR 20d NAILS.
  - USE GALVANIZED NAILS AT ALL PRESSURE TREATED WOOD SILL PLATES.
- MACHINE APPLIED NAILS:

USE OF MACHINE NAILING IS SUBJECT TO A SATISFACTORY JOB SITE DEMONSTRATION FOR EACH PROJECT AND THE APPROVAL BY THE STRUCTURAL ENGINEER AND THE DIVISION OF THE STATE ARCHITECT. THE APPROVAL IS SUBJECT TO CONTINUED SATISFACTORY PERFORMANCE. MACHINE NAILING WILL NOT BE APPROVED IN 5/16" PLYWOOD. IF NAIL HEADS PENETRATE THE OUTER PLY MORE THAN WOULD BE NORMAL FOR A HAND HAMMER OR IF MINIMUM ALLOWABLE EDGE DISTANCES ARE NOT MAINTAINED THE PERFORMANCE WILL BE DEEMED UNSATISFACTORY.